

# Solving Radical Equations and Checking Extraneous Roots

## Answer Key

### Algebra 2

#### Quick Answers

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|                |               |        |                 |
|----------------|---------------|--------|-----------------|
| 1. 28          | 4. 5, 0, 5, 2 | 7. , 6 | 10. 4, -1, 4, 4 |
| 2. 8           | 5. 100        | 8. 4   |                 |
| 3. -2, 3, 0, 4 | 6. 150        | 9. 9   |                 |

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1. Solve  $\sqrt{x-3} = 5$ .

**Solution:** the radical is already isolated, so square both sides:  $x - 3 = 25$ , hence  $x = 28$ . Check:  $\sqrt{28-3} = \sqrt{25} = 5$  ✓. Both sides were non-negative before squaring, so no extraneous root could appear.  $\boxed{x = 28}$

2. Rewrite and evaluate  $16^{3/4}$ .

**Solution:** a rational exponent is root-then-power:  $16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = 8$ . Taking the fourth root first keeps the arithmetic small; the other order needs  $16^3 = 4096$  before rooting.

$\boxed{8}$

3.  $\sqrt{x+6} = x$ , squared equation  $x^2 - x - 6 = 0$ .

(a)  $x^2 - x - 6 = (x-3)(x+2) = 0$  gives the candidates  $\boxed{x = -2}$  and  $\boxed{x = 3}$ .

(b) Substituting into  $\sqrt{x+6} - x$ : at  $x = 3$ :  $\sqrt{9} - 3 = 0$  ✓; at  $x = -2$ :  $\sqrt{4} - (-2) = 2 + 2 = \boxed{4} \neq 0$  ✗.

(c)  $x = -2$  is **extraneous** — it solves  $x+6 = x^2$  but makes the original read  $2 = -2$ , and the principal square root is never negative. Solution:  $\boxed{x = 3}$

4. Solve  $\sqrt{3x+1} = x-1$ .

(a) Squaring:  $3x+1 = x^2 - 2x + 1$ , so  $x^2 - 5x = 0$  and  $x(x-5) = 0$ , giving candidates  $\boxed{x = 0}$  and  $x = 5$ .

(b) At  $x = 0$ :  $\sqrt{3(0)+1} - (0-1) = 1+1 = \boxed{2} \neq 0$  ✗ — the left side is 1 while the right side is -1, so this candidate is extraneous. At  $x = 5$ :  $\sqrt{16} = 4$  and  $5-1 = 4$  ✓.

(c) Solution set:  $\boxed{x = 5}$

*Why it happened:* the right side  $x-1$  is negative at  $x = 0$ , and a square root can never equal a negative number.

5. Free fall:  $t = \sqrt{h/16}$  with  $t = 2.5$  seconds; find  $h$  in feet.

**Solution:** substitute and solve the radical equation:

$$2.5 = \sqrt{\frac{h}{16}} \implies 6.25 = \frac{h}{16} \implies h = 100$$

Check in the model:  $\sqrt{100/16} = \sqrt{6.25} = 2.5$  ✓. Both sides are non-negative here (a time and a principal root), so the candidate cannot be extraneous.  $\boxed{h = 100 \text{ ft}}$

**6.** Skid marks:  $s = \sqrt{24x}$  with  $s = 60$  miles per hour; find  $x$  in feet.

**Solution:**

$$60 = \sqrt{24x} \implies 3600 = 24x \implies x = 150$$

Check:  $\sqrt{24 \cdot 150} = \sqrt{3600} = 60 \checkmark$ .  $\boxed{x = 150 \text{ ft}}$

*Domain note:* only  $x \geq 0$  makes sense as a skid length, and the model's own square root already refuses negative inputs.

**7.** Solve  $\sqrt{x-2} = -3$ , or explain why there is no real solution.

**Solution:** squaring gives  $x - 2 = 9$ , so  $x = 11$  is the only candidate. Substituting into  $\sqrt{x-2} + 3$  at  $x = 11$ :  $\sqrt{9} + 3 = \boxed{6} \neq 0$ , i.e.  $\sqrt{9} = 3 \neq -3 \times$ .

The principal square root outputs only values  $\geq 0$ , so  $\sqrt{x-2} = -3$  is impossible for every real  $x$ ; the candidate is extraneous and the equation has  $\boxed{\text{no real solution}}$

This is the cleanest illustration of why the check is mandatory: squaring manufactured a root out of an equation with none.

**8.** Solve  $x^{3/2} = 8$ .

**Solution:** in radical form the equation is  $(\sqrt{x})^3 = 8$ . Take the cube root of both sides:  $\sqrt{x} = 2$ , then square:  $x = 4$ . Check:  $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8 \checkmark$ .  $\boxed{x = 4}$

**No negative solution:**  $x^{3/2}$  contains  $\sqrt{x}$ , which is undefined for  $x < 0$  in the real numbers — the domain rules negatives out before any solving happens.

**9.** Solve  $\sqrt{x+7} - \sqrt{x} = 1$ .

**Solution:** isolate one radical first, or squaring leaves two of them:

$$\begin{aligned} \sqrt{x+7} &= 1 + \sqrt{x} \\ x+7 &= 1 + 2\sqrt{x} + x && \text{(square both sides)} \\ 6 &= 2\sqrt{x} \implies \sqrt{x} = 3 && \text{(isolate the radical)} \\ x &= 9 && \text{(square again)} \end{aligned}$$

Check:  $\sqrt{9+7} - \sqrt{9} = 4 - 3 = 1 \checkmark$ .  $\boxed{x = 9}$

**10.**  $\sqrt{x+5} = x-1$  with the table of both sides.

(a) At  $x = -1$ :  $\sqrt{4} = 2$  but  $x - 1 = -2$  — *not equal*. At  $x = 4$ :  $\sqrt{9} = 3$  and  $x - 1 = 3$  — *equal*. (Squaring gives  $x + 5 = x^2 - 2x + 1$ , i.e.  $x^2 - 3x - 4 = 0$ , whose roots are the two candidates  $\boxed{x = -1}$  and  $x = 4$ .)

(b) Solution set:  $\boxed{x = 4}$

(c) The curve  $y = \sqrt{x+5}$  lies on or above the  $x$ -axis everywhere, so it can only meet  $y = x - 1$  where that line is non-negative, i.e. for  $x \geq 1$  — one intersection. Squaring replaces the line by the parabola  $y = (x-1)^2$ , which is non-negative on both sides of  $x = 1$ , so it also meets the graph of  $y = x + 5$  at  $x = -1$ . The table shows the cost: at  $x = -1$  the two original sides differ by  $\boxed{4}$ , so that intersection belongs to the squared equation only.